

پروژه مدار های الکتریکی (۱)

نام استاد : دکتر افضلیان

اعضای گروه : علیرضا منتجب ، عرفان فقهی ، علی وثوق نیا ، ارغوان معماری



فهرست

شماتیک

پاسخ ورودی صفر

فرکانس های طبیعی

پاسخ پله

پاسخ ضربه

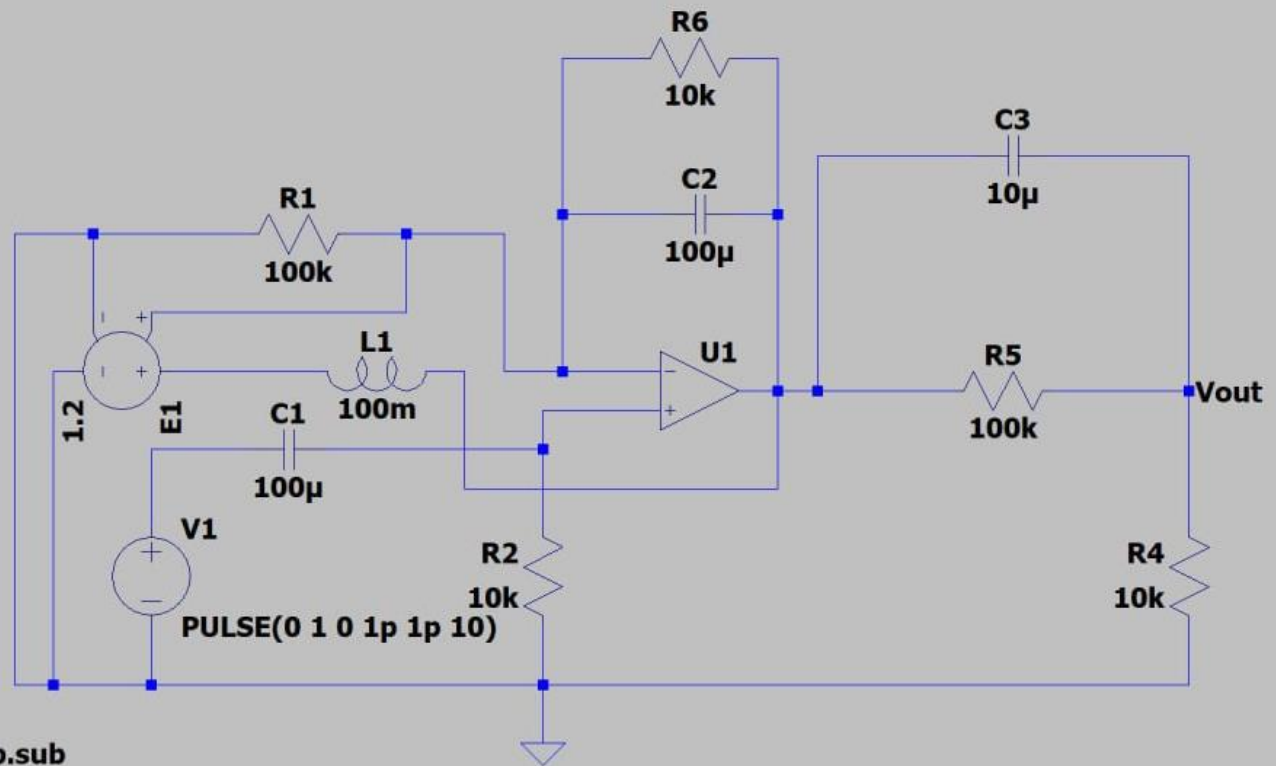
معادلات حالت

حالت دایمی سینوسی

امپدانس و ادمیتانس دیده شده از دو سر منبع

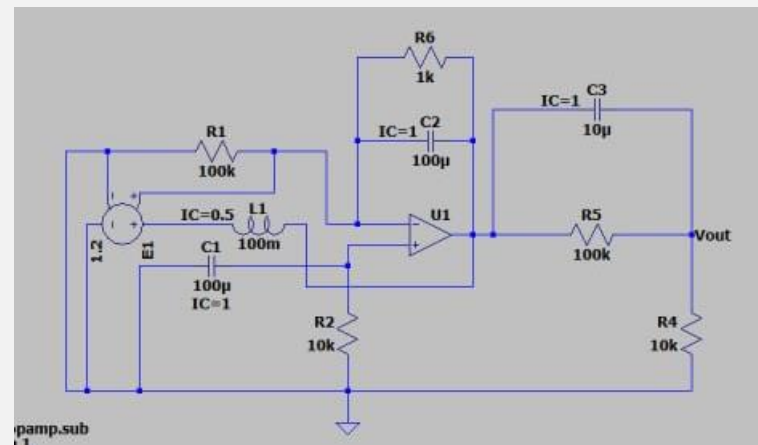
توان و بازدهی

شماتیک مدار



پاسخ ورودی صفر

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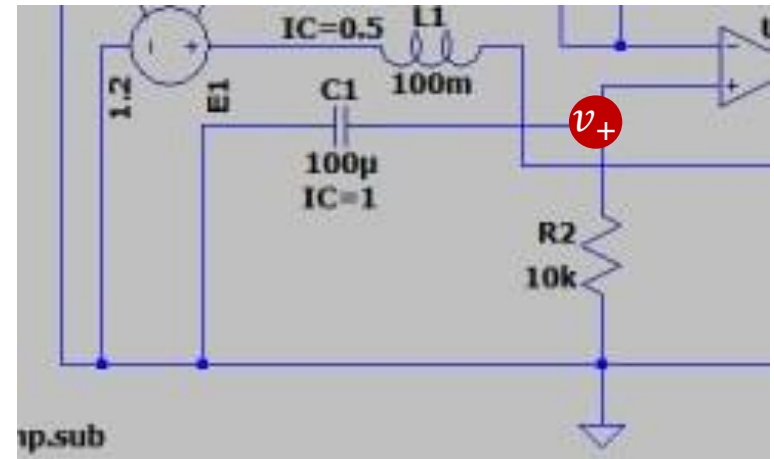
$$v_1(0) = v_{c_1}(0) = v_{c_2}(0) = 1v$$

$$c_1 \frac{dv_+}{dt} + \frac{v_+}{R_2} = 0$$

$$v_+ = k e^{-\frac{t}{R_2 c_1}}$$

$$K = v_{c_1}(0)$$

$$v_+ = e^{-\frac{t}{R_2 c_1}}$$



$$\frac{v_-}{R_1} + \frac{v_- - v_1}{R_6} + c_2 \frac{dv_- - v_1}{dt} = 0$$

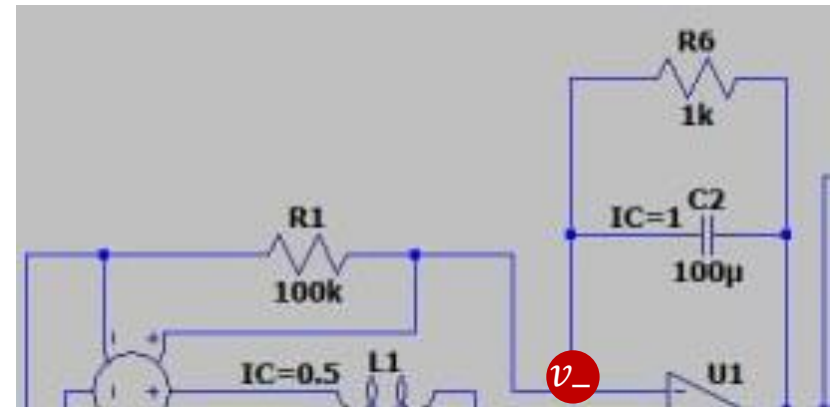
$$\left(\frac{v_{c_1}(0)}{c_2 R_1} + \frac{v_{c_1}(0)}{c_2 R_1} + \frac{v_{c_1}(0)}{c_1 R_2} \right) e^{-\frac{t}{R_2 c_1}} = \frac{dv_1}{dt} + \frac{v_1}{c_2 R_6}$$

$$v_1 = k_1 e^{-\frac{t}{R_6 c_2}} + A e^{-\frac{t}{R_2 c_1}}$$

$$A = \frac{\left(\frac{v_{c_1}(0)}{c_2 R_1} + \frac{v_{c_1}(0)}{c_2 R_1} + \frac{v_{c_1}(0)}{c_1 R_2} \right)}{\frac{1}{c_2 R_6} - \frac{1}{c_1 R_2}} = 0.03$$

$$v_1(0) = v_{c_1}(0) + v_{c_2}(0) = k_1 + A$$

$$k_1 = v_{c_1}(0) + v_{c_2}(0) - A = 1.97$$



$$v_1 = 1.97 e^{-\frac{t}{R_6 c_2}} + 0.03 e^{-\frac{t}{R_2 c_1}}$$

$$\frac{v_{out} - v_1}{R_5} + C_3 \frac{dv_{out} - v_1}{dt} + \frac{v_{out}}{R_4} = 0$$

$$\frac{v_{out}}{c_3 R_5} + \frac{dv_{out}}{dt} + \frac{v_{out}}{R_4 c_3} = \frac{v_1}{c_3 R_5} + \frac{dv_1}{dt}$$

$$v_{out} = B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} + C e^{-\frac{t}{R_6 c_2}} + D e^{-\frac{t}{R_2 c_1}}$$

$$v_{out}(0) = v_{c_1}(0) + v_{c_2}(0) + v_{c_3}(0) = 3 = B + C + D$$

$$e^{-\frac{t}{R_6 c_2}} \left(\frac{k_1}{c_3 R_5} + \frac{-k_1}{c_2 R_6} \right) + e^{-\frac{t}{R_2 c_1}} \left(\frac{A}{c_3 R_5} + \frac{-A}{c_1 R_2} \right) = \frac{v_{out}}{c_3 R_5} + \frac{dv_{out}}{dt} + \frac{v_{out}}{R_4 c_3}$$

$$e^{-\frac{t}{R_6 c_2}} \left(\frac{k_1}{c_3 R_5} + \frac{-k_1}{c_2 R_6} \right) + e^{-\frac{t}{R_2 c_1}} \left(\frac{A}{c_3 R_5} + \frac{-A}{c_1 R_2} \right) = \frac{C e^{-\frac{t}{R_6 c_2}} + D e^{-\frac{t}{R_2 c_1}}}{c_3 R_5} + \frac{-C}{c_2 R_6} e^{-\frac{t}{R_6 c_2}}$$

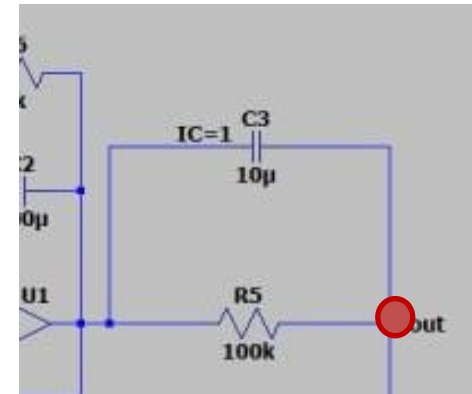
$$+ \frac{-D}{c_1 R_2} e^{-\frac{t}{R_2 c_1}} + \frac{C e^{-\frac{t}{R_6 c_2}} + D e^{-\frac{t}{R_2 c_1}}}{c_3 R_4}$$

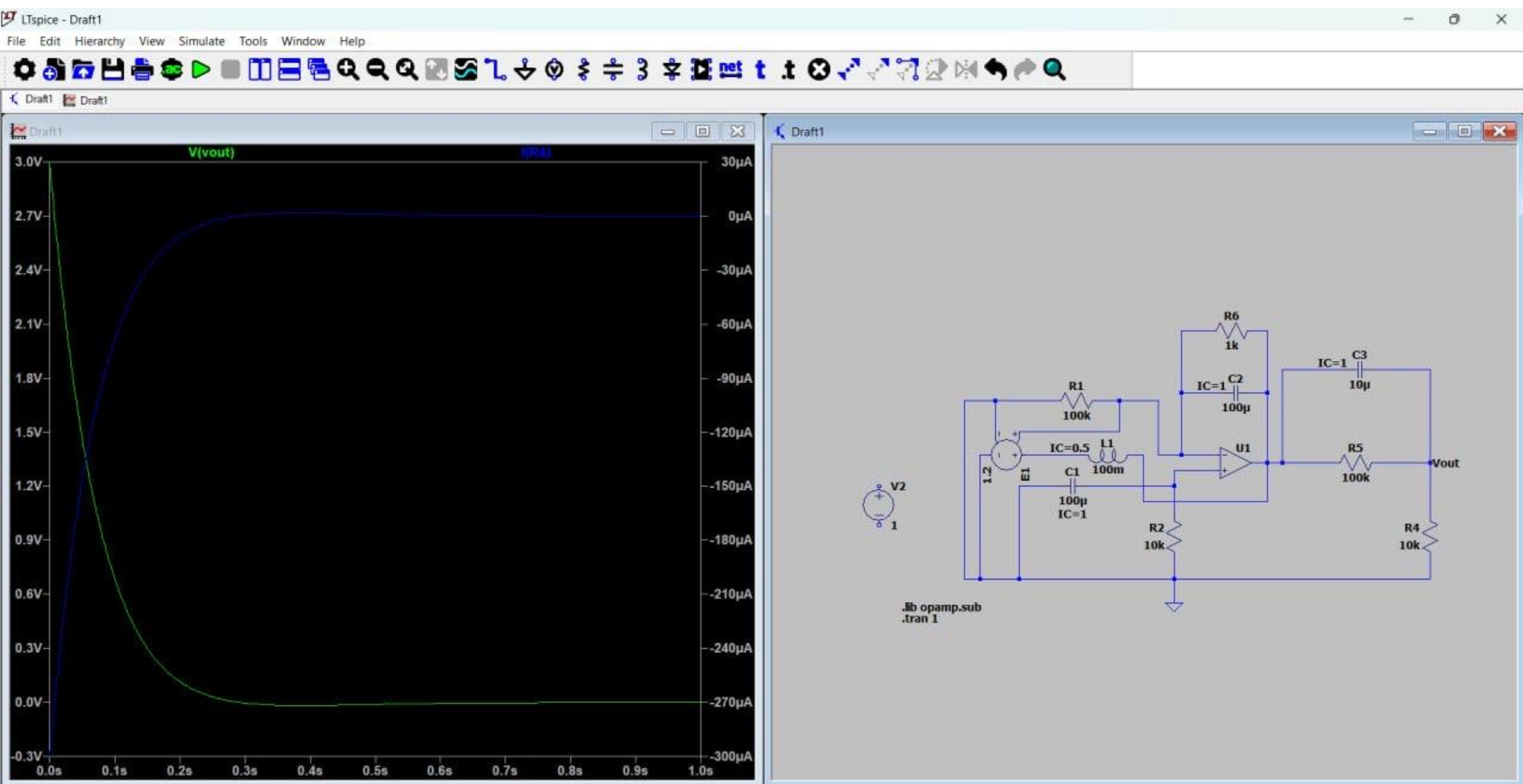
$$C = \frac{\frac{k_1}{c_3 R_5} + \frac{-k_1}{c_2 R_6}}{\frac{-1}{c_2 R_6} + \frac{1}{c_3 R_5} + \frac{1}{c_3 R_4}} = -0.2$$

$$D = \frac{\frac{A}{c_3 R_5} + \frac{-A}{c_1 R_2}}{\frac{-1}{c_1 R_2} + \frac{1}{c_3 R_5} + \frac{1}{c_3 R_4}} = -0.003$$

$$B = 3.203$$

$$v_{out} = 3.203 e^{-11t} - 0.2 e^{-10t} - 0.003 e^{-0.1t}$$

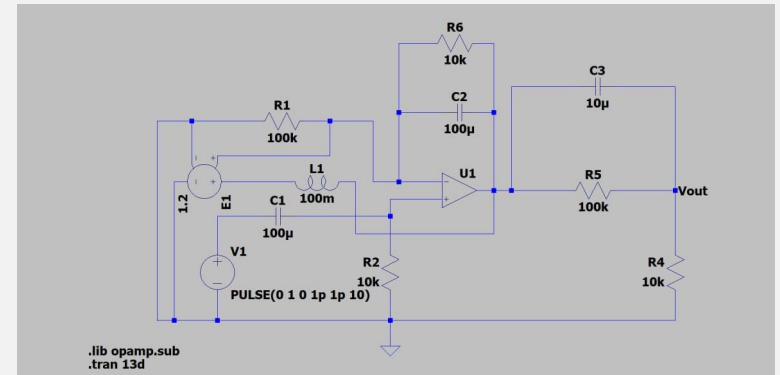




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پاسخ ورودی



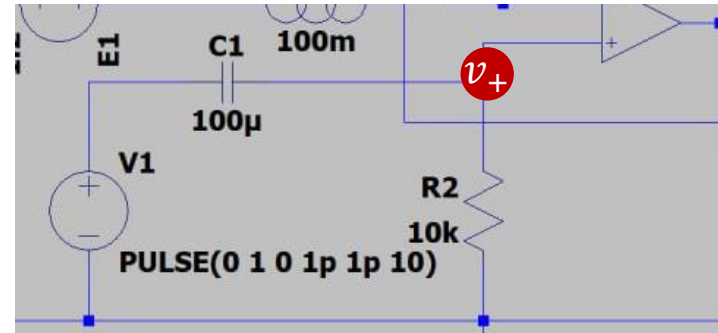
$$C_1 \frac{dv_+}{dt} + \frac{v_+}{R_2} = C_1 \frac{dv_{in}}{dt}$$

$$v_+ = ke^{-\frac{t}{R_2 C_1}} \cdot u(t)$$

$$\frac{-k}{R_2} e^{-\frac{t}{R_2 C_1}} + k C_1 \delta + \frac{k e^{-\frac{t}{R_2 C_1}} \cdot u(t)}{R_2} = C_1 \delta$$

$$k=1$$

$$v_+ = e^{-\frac{t}{R_2 C_1}} \cdot u(t)$$

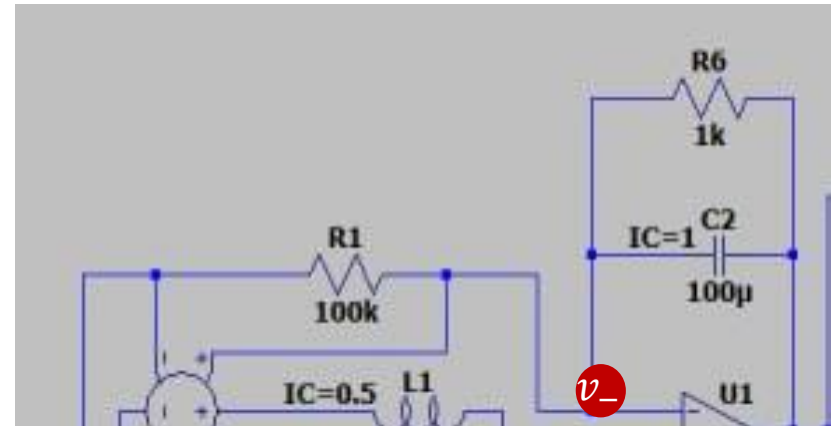


$$\frac{v_-}{R_1} + C_2 \frac{dv_-}{dt} + \frac{v_-}{R_6} = C_2 \frac{dv_1}{dt} + \frac{v_1}{R_6} \quad v_- = v_+$$

$$v_1 = Ae^{-\frac{t}{R_2 C_1}} \cdot u(t) + Be^{-\frac{t}{R_6 C_2}} \cdot u(t)$$

$$\frac{e^{-\frac{t}{R_2 C_1}} \cdot u(t)}{R_1} - \frac{C_2}{C_1 R_2} e^{-\frac{t}{R_2 C_1}} \cdot u(t) + \frac{e^{-\frac{t}{R_2 C_1}} \cdot u(t)}{R_6} + C_2 \delta = \frac{-A C_2}{C_1 R_2} e^{-\frac{t}{R_2 C_1}} \cdot u(t) + \frac{-B}{R_6} e^{-\frac{t}{R_6 C_2}} \cdot u(t) + A C_2 \delta +$$

$$B C_2 \delta + \frac{A e^{-\frac{t}{R_2 C_1}} \cdot u(t)}{R_6} + \frac{B e^{-\frac{t}{R_6 C_2}} \cdot u(t)}{R_6}$$



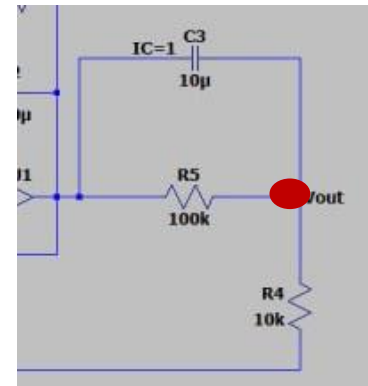
$$\frac{v_{out} - v_1}{R_5} + C_3 \frac{dv_{out} - v_1}{dt} + \frac{v_{out}}{R_4} = 0$$

$$\frac{v_{out}}{c_3 R_5} + \frac{dv_{out}}{dt} + \frac{v_{out}}{R_4 c_3} = \frac{v_1}{c_3 R_5} + \frac{dv_1}{dt}$$

$$v_{out} = B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} \cdot u(t) + C e^{-\frac{t}{R_6 c_2}} \cdot u(t) + D e^{-\frac{t}{R_2 c_1}} \cdot u(t)$$

$$\frac{B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} \cdot u(t) + C e^{-\frac{t}{R_6 c_2}} \cdot u(t) + D e^{-\frac{t}{R_2 c_1}} \cdot u(t)}{c_3 R_5}$$

جمله اول



$$\frac{dv_{out}}{dt} = -\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3} B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} \cdot u(t) + B \cdot \delta - \frac{t}{R_6 c_2} C e^{-\frac{t}{R_6 c_2}} \cdot u(t) + C \cdot \delta - \frac{t}{R_2 c_1} D e^{-\frac{t}{R_2 c_1}} \cdot u(t) + D \cdot \delta$$

(جمله دوم)

$$\frac{B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} \cdot u(t) + C e^{-\frac{t}{R_6 c_2}} \cdot u(t) + D e^{-\frac{t}{R_2 c_1}} \cdot u(t)}{c_3 R_4} - \frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3} B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} \cdot u(t) + B \cdot \delta - \frac{t}{R_6 c_2} C e^{-\frac{t}{R_6 c_2}} \cdot u(t)$$

$$+ C \cdot \delta - \frac{t}{R_2 c_1} D e^{-\frac{t}{R_2 c_1}} \cdot u(t) + D \cdot \delta + \frac{B e^{-\frac{t(c_3 R_5 + R_4 c_2)}{c_3 R_5 R_4 c_3}} \cdot u(t) + C e^{-\frac{t}{R_6 c_2}} \cdot u(t) + D e^{-\frac{t}{R_2 c_1}} \cdot u(t)}{c_3 R_5} =$$

$$\frac{1.01 e^{-\frac{t}{R_2 c_1}} \cdot u(t) - 0.01 e^{-\frac{t}{R_6 c_2}} \cdot u(t)}{c_3 R_5} - \frac{1.01}{R_2 c_1} e^{-\frac{t}{R_2 c_1}} \cdot u(t) + 1.01 \delta + \frac{0.01}{R_6 c_2} e^{-\frac{t}{R_6 c_2}} \cdot u(t) - 0.01 \delta$$

$$\frac{C}{c_3 R_4} - \frac{C}{c_2 R_6} + \frac{C}{c_3 R_5} = \frac{-0.01}{c_3 R_5} + \frac{0.01}{R_6 c_2}$$

$$c = 0.0001$$

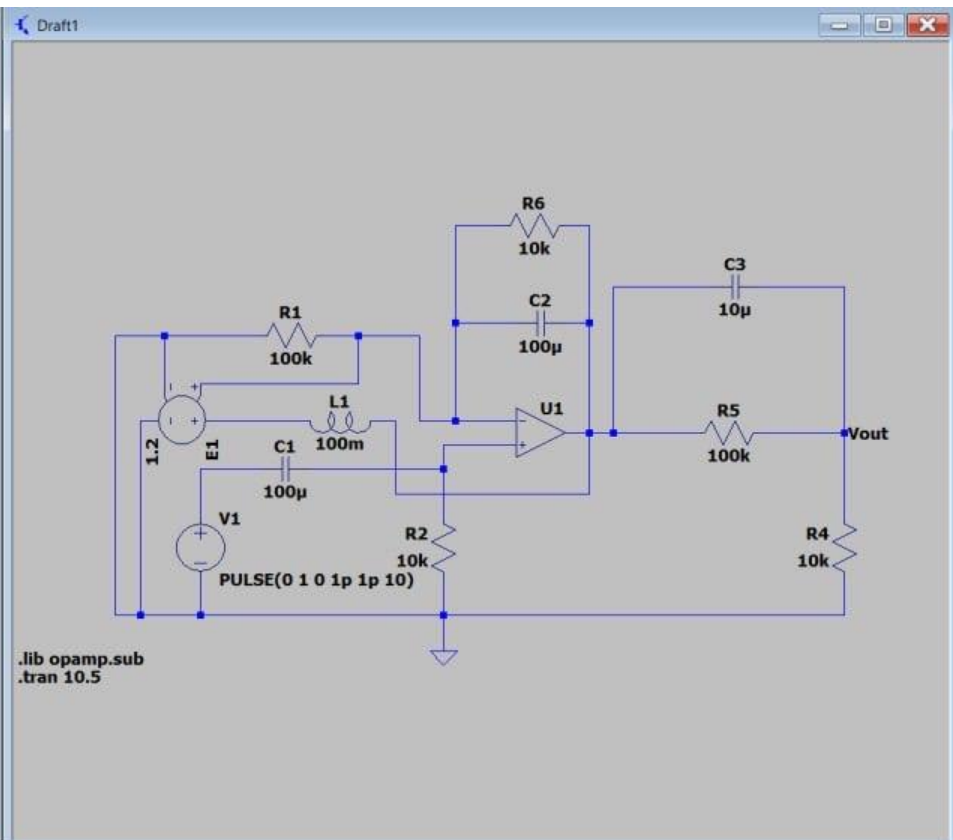
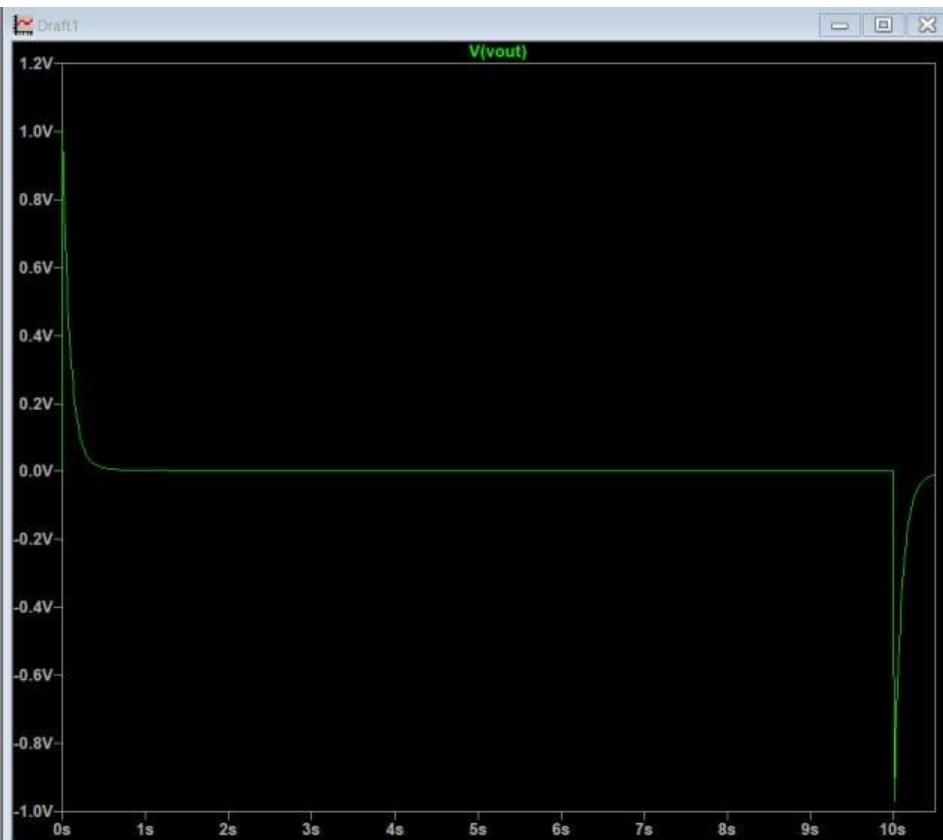
$$\frac{D}{c_3 R_4} - \frac{D}{c_1 R_2} + \frac{D}{c_3 R_5} = \frac{1.01}{c_3 R_5} - \frac{1.01}{R_6 c_2}$$

$$B = -0.1$$

$$B + C + D = 1$$

$$D = 1.0999$$

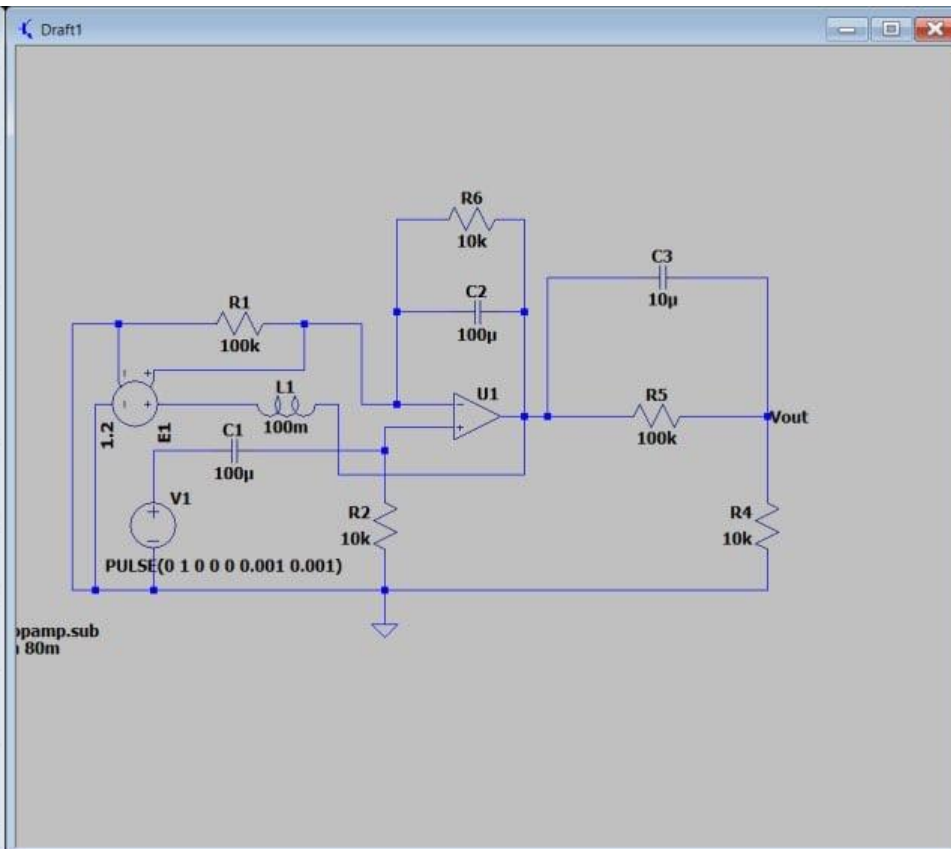
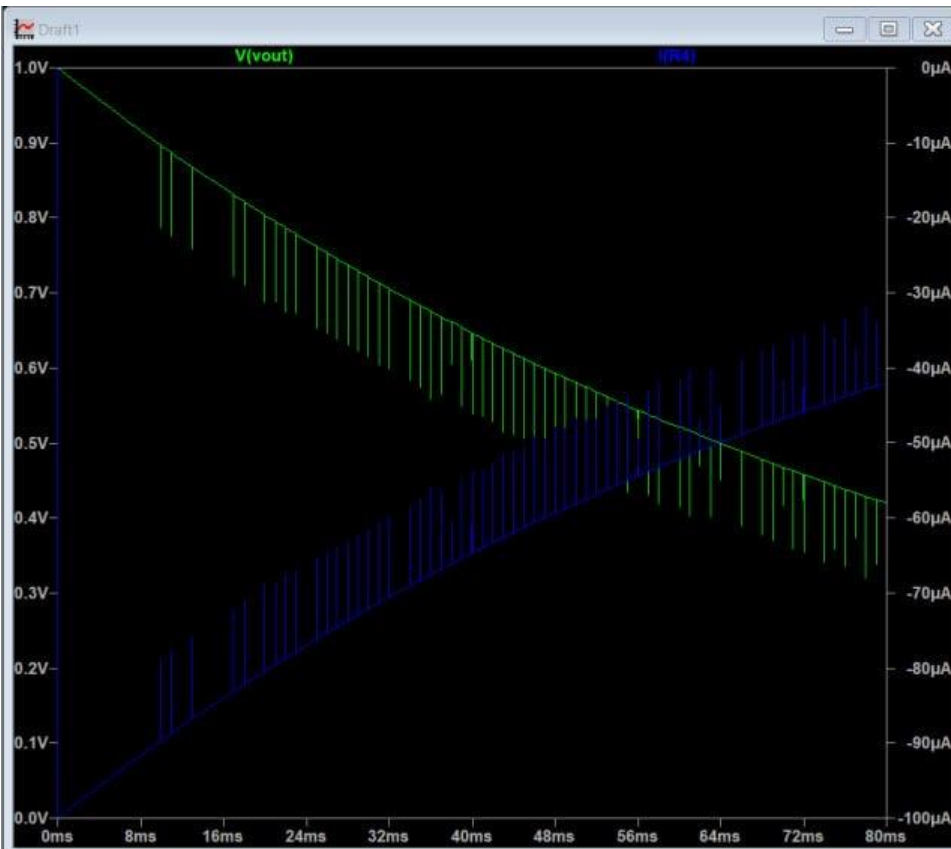
$$v_{out} = -0.1 e^{-11t} \cdot u(t) + 0.0001 e^{-10t} \cdot u(t) + 1.0999 e^{-0.1t} \cdot u(t)$$



پاسخ ضربه

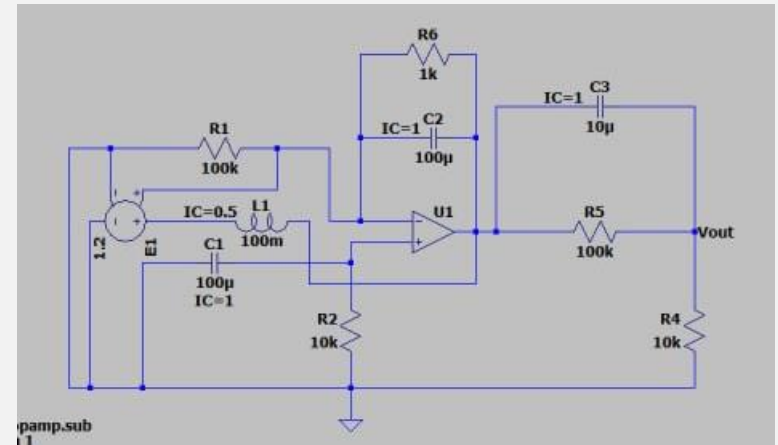
$$\frac{dv_{out}}{dt} = 1.1e^{-11t} \cdot u(t) + -0.1 \cdot \delta - 0.001e^{-10t} \cdot u(t) + 0.0001 \cdot \delta - 0.1099e^{-0.1t} \cdot u(t) + 1.0999 \cdot \delta$$

$$\frac{dv_{out}}{dt} = 1.1e^{-11t} \cdot u(t) - 0.001e^{-10t} \cdot u(t) - 0.1099e^{-0.1t} \cdot u(t) + \delta$$



معادلات حالت

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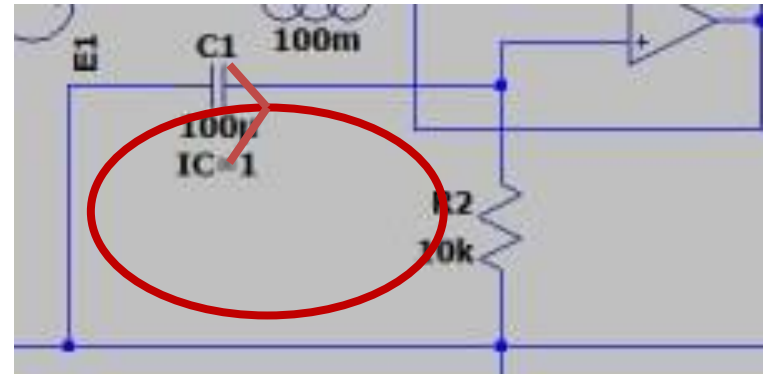


$$-v_{C_1} + v_{R_2} = 0$$

$$v_+ = v_- = v_{R_2} = v_{C_1} = R_2 I_{C_1} = C_1 R_2 \frac{dv_{C_1}}{dt}$$

$$v_{C_1} = C_1 R_2 \frac{dv_{C_1}}{dt}$$

$$\frac{dv_{C_1}}{dt} = \frac{v_{C_1}}{C_1 R_2}$$

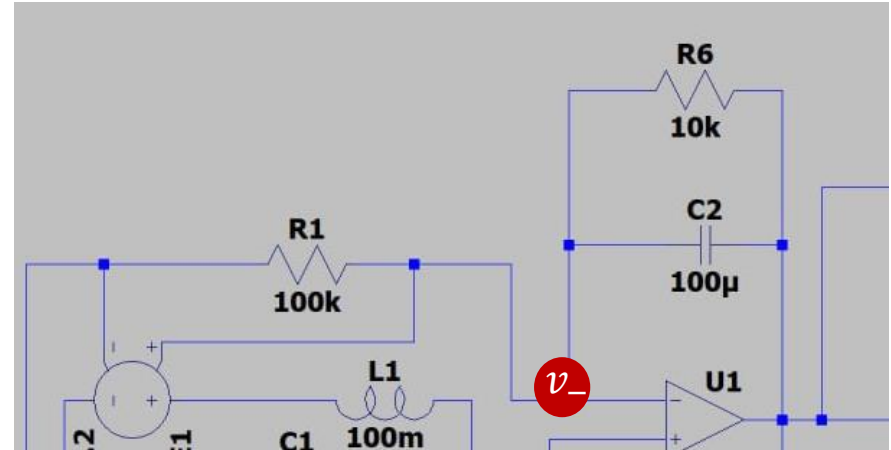


$$I_{R_1} + I_{C_2} + I_{R_6} = 0$$

$$\frac{v_-}{R_1} + C_2 \frac{dv_{C_2}}{dt} + \frac{v_{C_2}}{R_6} = 0$$

$$\frac{v_{C_1}}{R_1} + C_2 \frac{dv_{C_2}}{dt} + \frac{v_{C_2}}{R_6} = 0$$

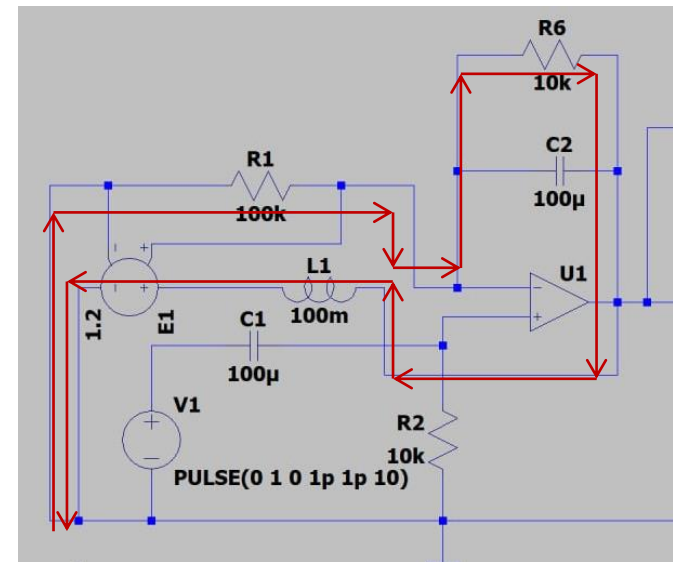
$$\frac{dv_{C_2}}{dt} = -\frac{v_{C_1}}{R_1 C_2} - \frac{v_{C_2}}{R_6 C_2}$$



$$v_L + v_{E_1} - v_{R_1} + v_{R_6} = 0$$

$$L \frac{di_L}{dt} + v_{c_2} + 0.2v_{c_1} = 0$$

$$\frac{di_L}{dt} = -\frac{v_{c_2}}{L} - \frac{0.2v_{c_1}}{L}$$



$$I_{R_4} + I_{C_3} + I_{R_5} = 0$$

$$\frac{v_{out}}{R_4} + C_3 \frac{dv_{c_3}}{dt} + \frac{v_{c_3}}{R_5} = 0$$

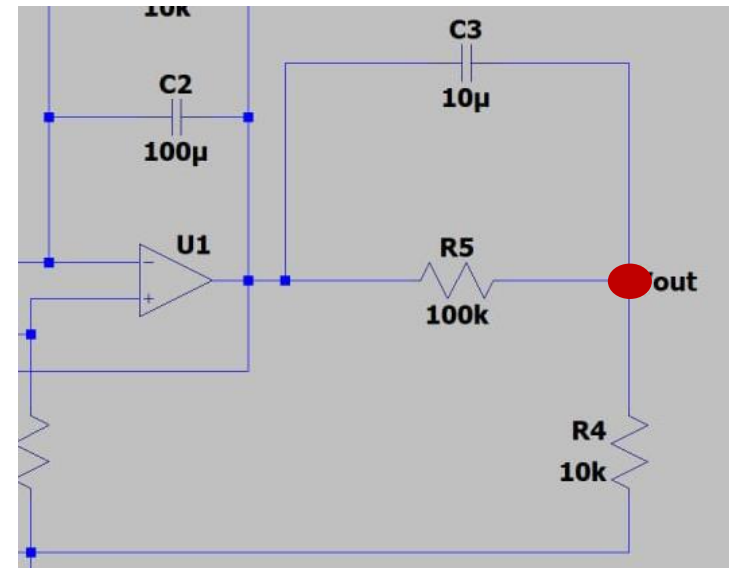
$$v_{out} + v_{c_3} = v_1$$

$$v_1 - v_{c_2} = v_-$$

$$v_{out} = v_{c_2} + v_- - v_{c_3} = v_{c_2} + v_{c_1} - v_{c_3}$$

$$\frac{dv_{c_3}}{dt} = -\frac{v_{out}}{R_4 c_3} - \frac{v_{c_3}}{R_5 c_3}$$

$$\frac{dv_{c_3}}{dt} = -\frac{v_{c_1}}{R_4 c_3} - \frac{v_{c_2}}{R_4 c_3} + \left(\frac{1}{R_4 c_3} - \frac{1}{R_5 c_3}\right)v_{c_3}$$



$$\frac{dv_{c_1}}{dt} = \frac{v_{c_1}}{C_1 R_2}$$

$$\frac{dv_{c_2}}{dt} = -\frac{v_{c_1}}{R_1 C_2} - \frac{v_{c_2}}{R_6 C_2}$$

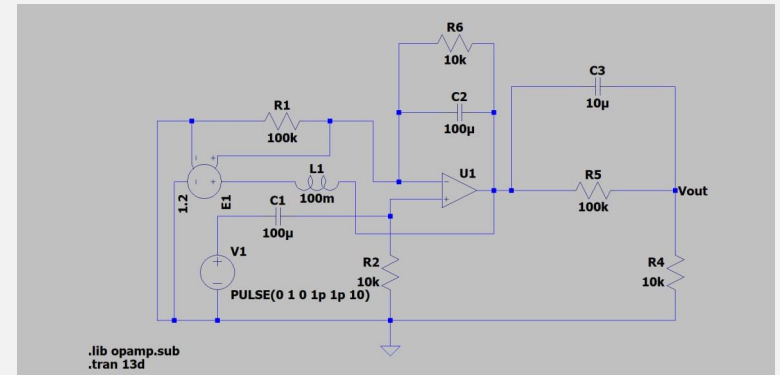
$$\frac{dv_{c_3}}{dt} = -\frac{v_{c_1}}{R_4 C_3} - \frac{v_{c_2}}{R_4 C_3} + \left(\frac{1}{R_4 C_3} - \frac{1}{R_5 C_3}\right)v_{c_3}$$

$$\frac{di_L}{dt} = -\frac{0.2v_{c_1}}{L} - \frac{v_{c_2}}{L}$$

$$\begin{bmatrix} \frac{dv_{c_1}}{dt} \\ \frac{dv_{c_2}}{dt} \\ \frac{dv_{c_3}}{dt} \\ \frac{di_L}{dt} \end{bmatrix} = \begin{bmatrix} \frac{v_{c_1}}{C_1 R_2} & 0 & 0 & 0 \\ -\frac{v_{c_1}}{R_1 C_2} & -\frac{v_{c_2}}{R_6 C_2} & 0 & 0 \\ -\frac{1}{R_4 C_3} & -\frac{1}{R_4 C_3} & \left(\frac{1}{R_4 C_3} - \frac{1}{R_5 C_3}\right) & 0 \\ -\frac{0.2v_{c_1}}{L} & -\frac{v_{c_2}}{L} & 0 & 0 \end{bmatrix} \begin{bmatrix} v_{c_1} \\ v_{c_2} \\ v_{c_3} \\ i_L \end{bmatrix}$$

حالت دایمی سینوسی

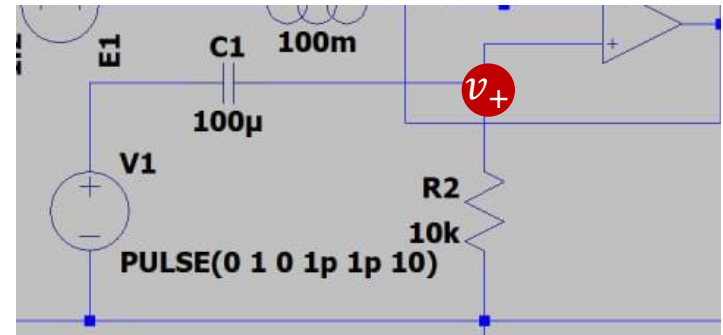
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$$\frac{v_+ - E}{\frac{1}{j\omega c_1}} + \frac{v_+}{R_2} = 0$$

$$(j\omega c_1 + \frac{1}{R_2})v_+ = j\omega C_1 E$$

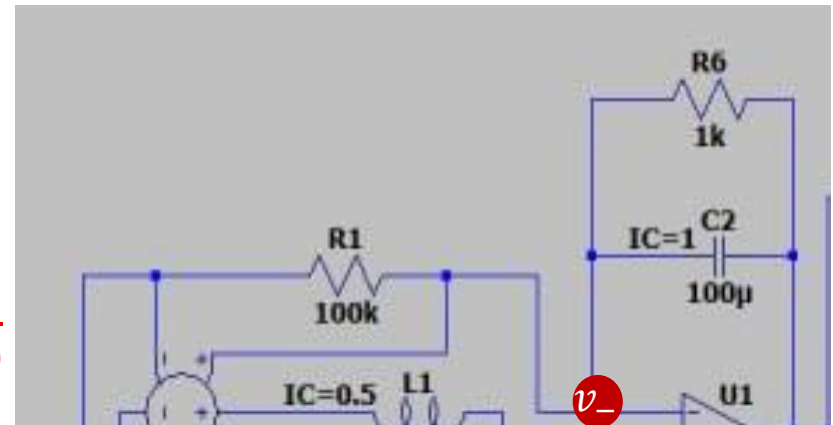
$$V_+ = \frac{j\omega C_1 E}{(j\omega c_1 + \frac{1}{R_2})}$$



$$\frac{v_-}{R_1} + \frac{v_- - v_1}{R_6} + \frac{v_- - v_1}{\frac{1}{j\omega c_1}} = 0$$

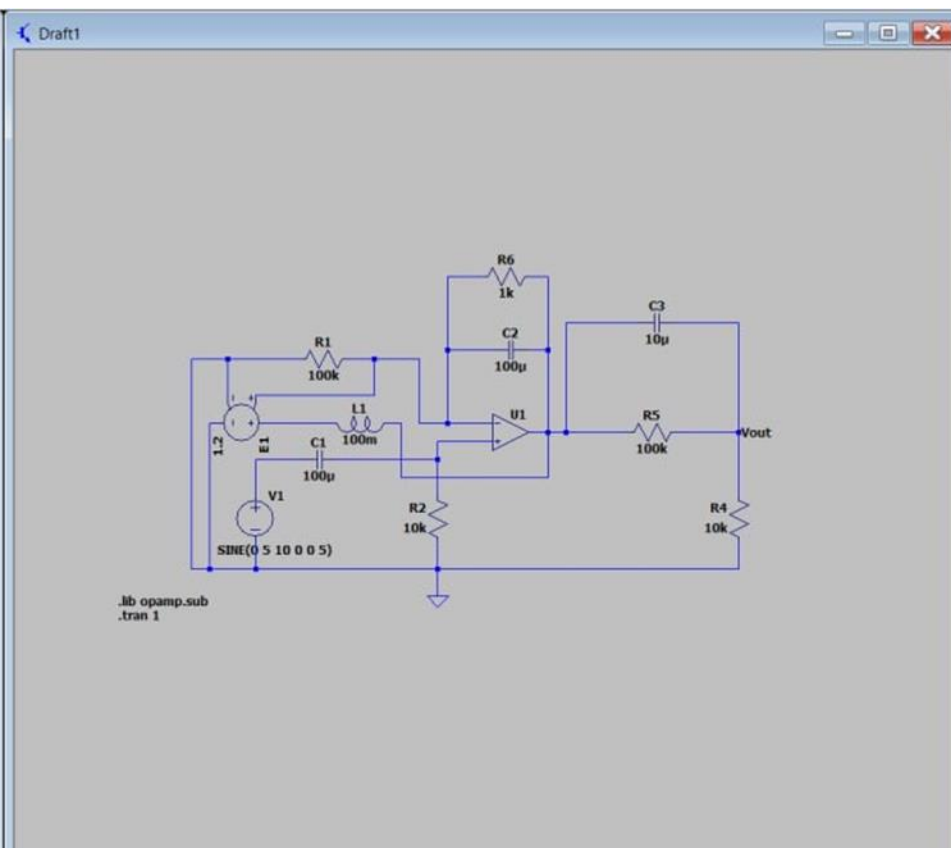
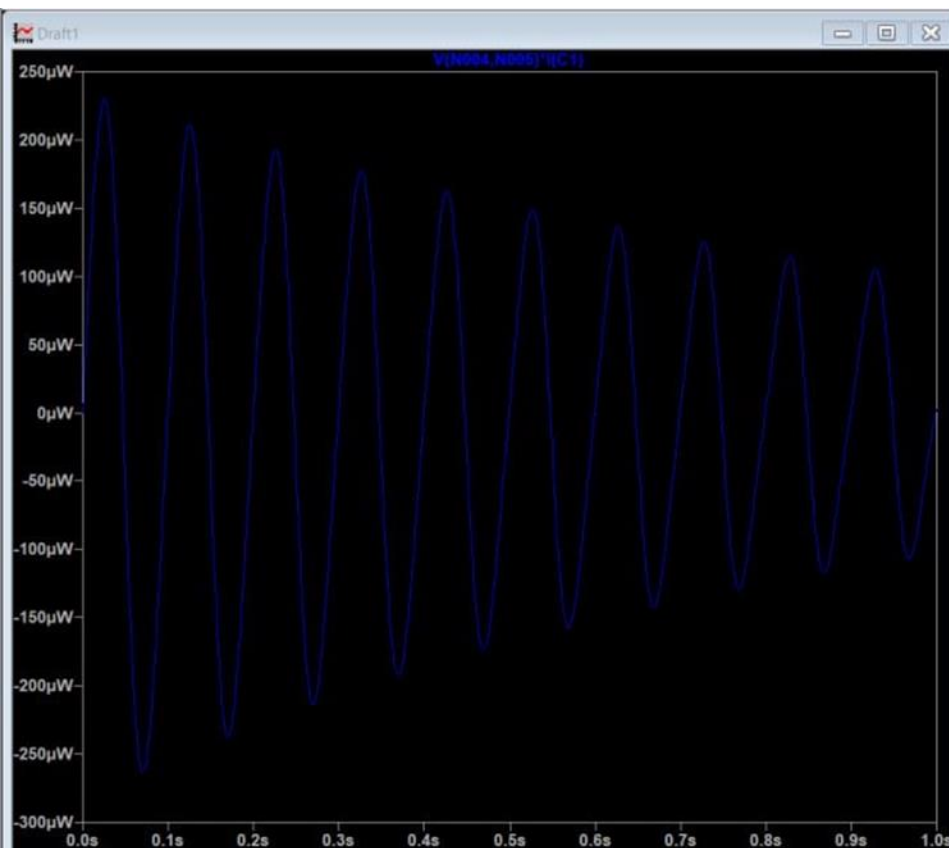
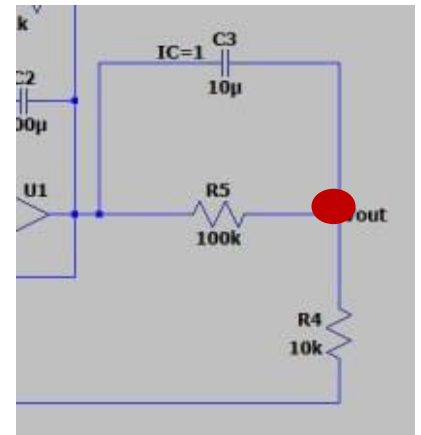
$$V_+ \left(\frac{1}{R_6} + \frac{1}{R_1} + j\omega C_2 \right) = \left(\frac{1}{R_6} + j\omega C_2 \right) v_1$$

$$v_1 = \frac{\left(\frac{1}{R_6} + \frac{1}{R_1} + j\omega C_2 \right)}{\left(\frac{1}{R_6} + j\omega C_2 \right)} V_+ = \frac{\left(\frac{1}{R_6} + \frac{1}{R_1} + j\omega C_2 \right)}{\left(\frac{1}{R_6} + j\omega C_2 \right)} \frac{j\omega C_1 E}{(j\omega c_1 + \frac{1}{R_2})}$$



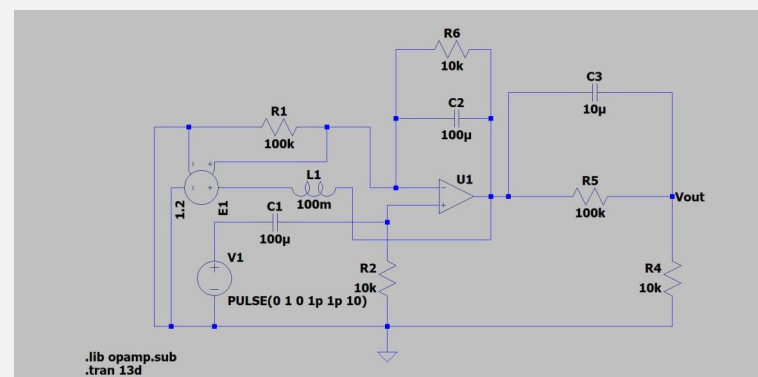
$$\frac{v_{out}-v_1}{R_5} + J\omega C_3(v_{out}-v_1) + \frac{v_{out}}{R_4} = 0$$

$$v_{out} = \frac{\left(\frac{1}{R_5} + J\omega C_3\right) v_1}{\left(\frac{1}{R_5} + J\omega C_3 + \frac{1}{R_4}\right)}$$



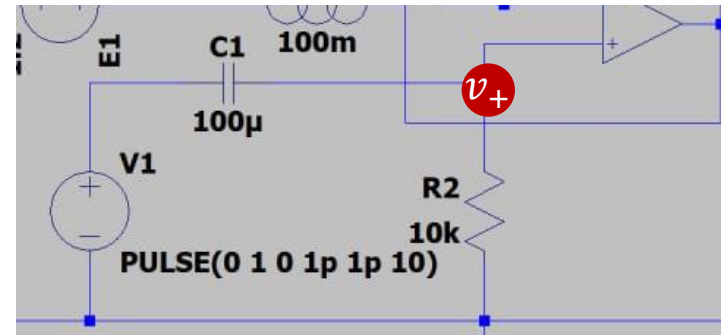
امپدانس و ادمیتانس دیده شده از دو سر منبع

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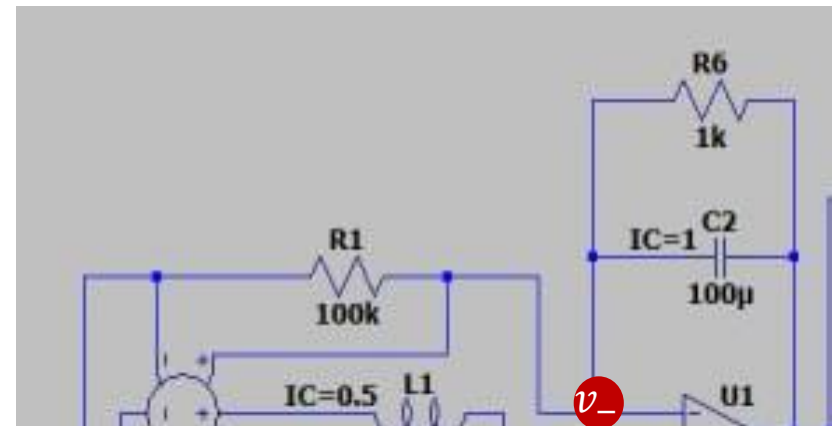
$$I_T = \frac{v_+}{R_2} \quad \frac{v_+ - v_T}{\frac{1}{j\omega c_1}} + I_T = 0$$

$$v_+ = v_T - \frac{I_T}{j\omega c_1}$$



$$\frac{v_-}{R_1} + \frac{v_- - v_1}{R_6} + \frac{v_- - v_1}{\frac{1}{j\omega c_2}} = 0$$

$$v_1 = \frac{(\frac{1}{R_1} + \frac{1}{R_6} + j\omega c_2)(v_T - \frac{I_T}{j\omega c_1})}{(\frac{1}{R_6} + j\omega c_2)}$$



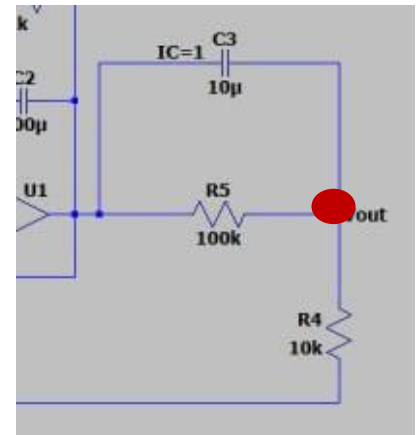
$$\frac{v_{out}-v_1}{R_5} + j\omega C_3(v_{out}-v_1) + \frac{v_{out}}{R_4} = 0$$

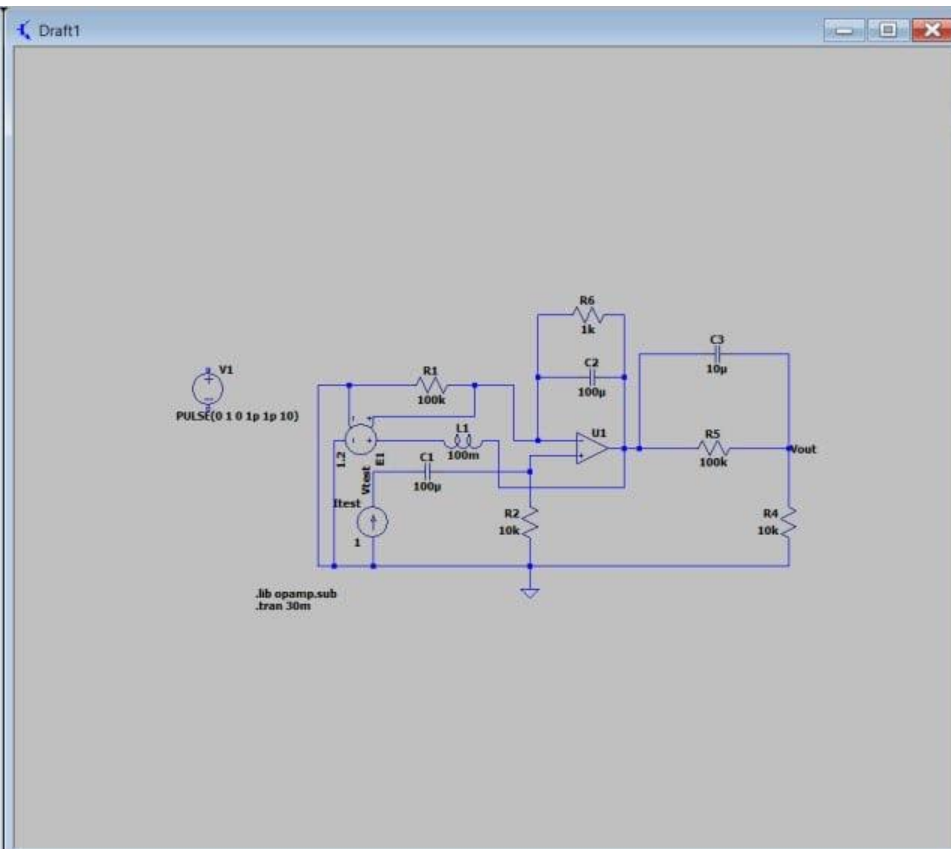
$$v_{out} = \frac{\left(\frac{1}{R_5} + j\omega C_3\right) v_1}{\left(\frac{1}{R_5} + j\omega C_3 + \frac{1}{R_4}\right)}$$

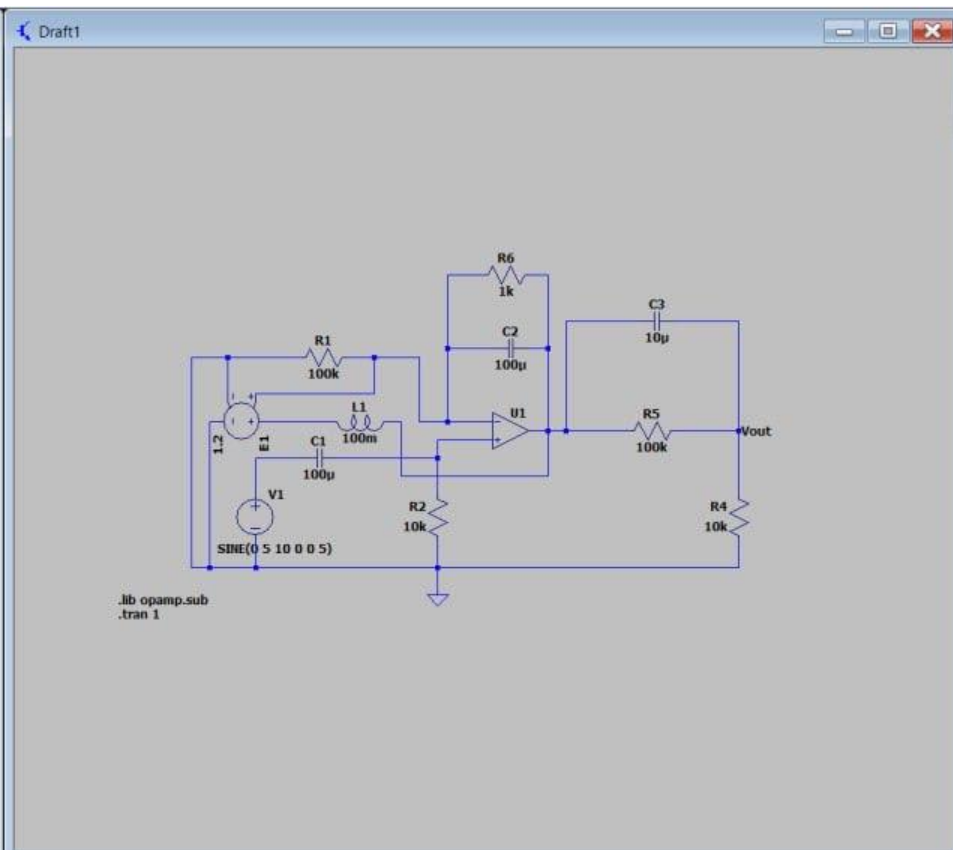
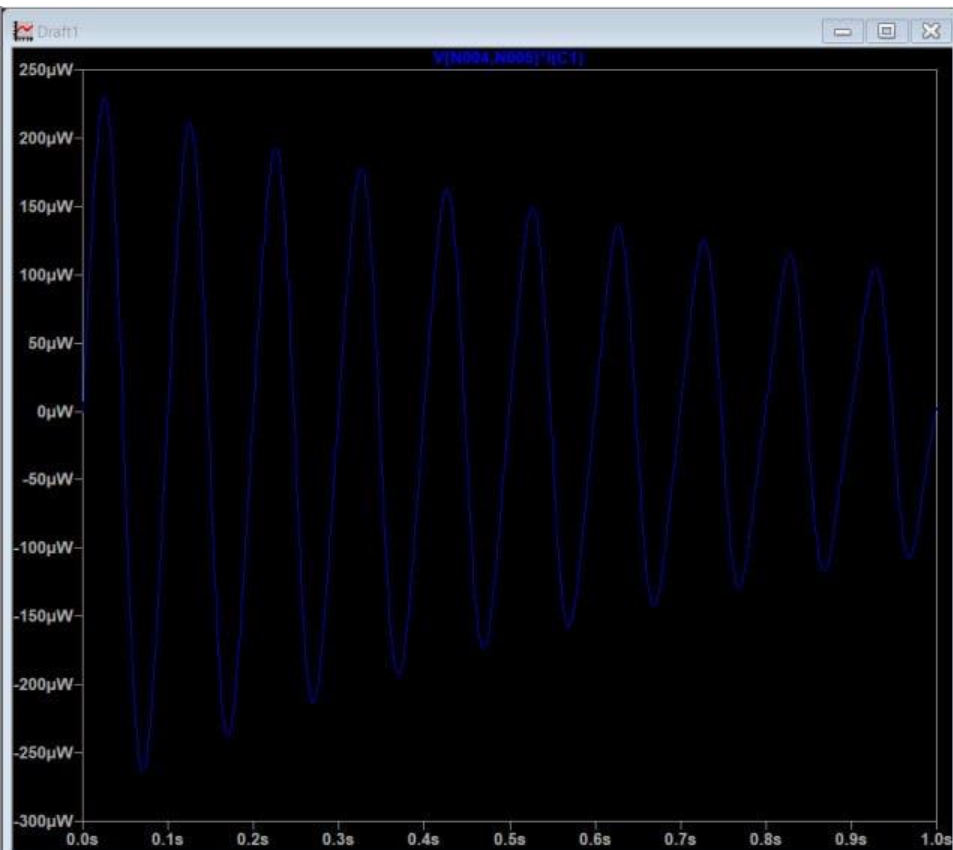
$$\frac{\left(\frac{1}{R_5} + j\omega C_3\right) \frac{\left(\frac{1}{R_1} + \frac{1}{R_6} + j\omega C_2\right) \left(v_T - \frac{I_T}{j\omega C_1}\right)}{\left(\frac{1}{R_6} + j\omega C_2\right)}}{\left(\frac{1}{R_5} + j\omega C_3 + \frac{1}{R_4}\right)} = \frac{\left(\frac{1}{R_5} + j\omega C_3\right) \frac{\left(\frac{1}{R_6} + \frac{1}{R_1} + j\omega C_2\right) j\omega C_1 v_T}{\left(\frac{1}{R_6} + j\omega C_2\right) \left(j\omega C_1 + \frac{1}{R_2}\right)}}{\left(\frac{1}{R_5} + j\omega C_3 + \frac{1}{R_4}\right)}$$

$$v_T - \frac{I_T}{j\omega C_1} = \frac{j\omega C_1 v_T}{\left(j\omega C_1 + \frac{1}{R_2}\right)}$$

$$v_T = \frac{I_T}{(1-j\omega C_1)(j\omega C_1)} + \frac{1}{(1-j\omega C_1)\left(j\omega C_1 + \frac{1}{R_2}\right)}$$







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